

Python Programming Fundamentals

List Basics

Programming Fundamentals Team

SETU

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Outline

1. Creating Lists

2. Indexing

3. Updating Items

4. Length and Membership

5. Common Error: IndexError



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1. Creating Lists

A list stores multiple values in one variable.

```
names = ["Ava", "Ben", "Cara"]
numbers = [10, 20, 30, 40]
mixed = ["Ava", 19, True]
empty = []
```



1. Creating Lists

A list stores multiple values in one variable.

```
names = ["Ava", "Ben", "Cara"]
numbers = [10, 20, 30, 40]
mixed = ["Ava", 19, True]
empty = []
```

Although Python allows mixed types, beginners should usually keep a list focused:

```
grades = [65, 72, 81, 90]
```



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2. Indexing

Each item has a position called an **index**.

```
names = ["Ava", "Ben", "Cara", "Dylan"]
#          0      1       2        3

print(names[0])   # Ava
print(names[2])   # Cara
```



2. Indexing

Each item has a position called an **index**.

```
names = ["Ava", "Ben", "Cara", "Dylan"]
#          0      1       2        3

print(names[0])   # Ava
print(names[2])   # Cara
```

Python also supports negative indexing.

```
print(names[-1])  # Dylan
print(names[-2])  # Cara
```



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3. Updating Items

Lists are **mutable**, which means they can be changed.

```
names = ["Ava", "Ben", "Cara"]

names[1] = "Brendan"

print(names)
```



3. Updating Items

Lists are **mutable**, which means they can be changed.

```
names = ["Ava", "Ben", "Cara"]

names[1] = "Brendan"

print(names)
```

Output:

```
['Ava', 'Brendan', 'Cara']
```



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4. Length and Membership

```
names = ["Ava", "Ben", "Cara"]

print(len(names)) #3
print("Ben" in names) #true
print("Zara" not in names) #true
```



4. Length and Membership

```
names = ["Ava", "Ben", "Cara"]

print(len(names)) #3
print("Ben" in names) #true
print("Zara" not in names) #true
```

Useful questions:

```
if "Ben" in names:
    print("Ben is in the class list")
```



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5. Common Error: IndexError

```
names = ["Ava", "Ben", "Cara"]
print(names[3])
```



5. Common Error: IndexError

```
names = ["Ava", "Ben", "Cara"]
print(names[3])
```

There is no index 3.

Valid indexes are 0, 1, and 2.

```
IndexError: list index out of range
```



Thanks for Watching - Any questions?

